MS PROJECT 4 (MTH700)
Matching Theory

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Professor, Department of Mathematics and Statistics

Declaration

I hereby declare that the work presented in the project report entitled "*Matching Theory*" is based on my own study, implementation, and analysis conducted under the supervision of Professor Saumyarup Sadhukhan.
Where ideas, definitions, or results are drawn from existing literature, proper references have been provided. The computational simulations and empirical analysis presented in this report are my own original work.

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Abstract

This project studies allocation mechanisms in matching theory under network constraints, focusing on the YRMH-IGYT mechanism over Tree-Structured Social Networks (TSSNs). While classical housing market models assume centralized participation and unrestricted exchange, many real-world allocation environments operate through hierarchical or referral-based networks.

YRMH-IGYT for TSSNs is known to satisfy desirable incentive and stability properties, including strategy-proofness, individual rationality, weak non-wastefulness, and strict core stability for neighbors (SC4N). However, global efficiency properties are significantly affected by network constraints. We provide a comprehensive theoretical and computational analysis of efficiency. First, we show that YRMH-IGYT fails to satisfy pair efficiency and Pareto efficiency in general. Monte Carlo simulations demonstrate that the probability of inefficiency increases with market size and approaches one.

To understand the source of inefficiency, we identify structural causes, particularly sibling-level conflicts and non-local interactions. We prove an impossibility result showing that no mechanism can simultaneously satisfy locality, SC4N, and Pareto efficiency on trees.

We then explore mechanisms that relax constraints to improve efficiency. A global cycle mechanism achieves Pareto efficiency by allowing mediator-based swaps, while a sibling swap mechanism provides a minimal local improvement. Empirical results show that Pareto gains increase with market size and grow at least logarithmically. Furthermore, we evaluate alternative decentralized protocols, starting with the Swap with Neighbors (SWN) mechanism. We formally prove that while SWN is individually rational, it is highly vulnerable to strategic manipulation and fails global efficiency due to intermediary bottlenecks. To resolve these limitations, we analyze the dynamic "Leave and Share" (LS) mechanism. Through theoretical proofs and computational verification, we demonstrate that by dynamically updating the network topology, LS restores strategy-proofness and successfully guarantees dynamic **Weak Pairwise Efficiency**.

These findings highlight a fundamental trade-off between incentives, stability, and efficiency in network-constrained matching markets.

1 Introduction

Matching theory studies the allocation of indivisible goods among agents with heterogeneous preferences. In the classical housing market model, mechanisms such as Top Trading Cycles (TTC) and YRMH-IGYT achieve strong guarantees, including strategy-proofness, Pareto efficiency, and strict core stability.

Recent work extends these models to settings where agents are embedded in social networks. In such environments, feasible exchanges are constrained by the underlying network structure. In particular, tree-structured social networks (TSSNs) model hierarchical systems where agents interact through parent-child relationships.

The YRMH-IGYT mechanism has been adapted to TSSNs and is known to satisfy several desirable properties, including strategy-proofness (SP), individual rationality (IR), weak group strategy-proofness, weak non-wastefulness, and strict core stability for neighbors (SC4N). These properties ensure incentive compatibility and local stability.

However, global efficiency properties are less understood in network-constrained settings. This project investigates the following fundamental questions:

- Does YRMH-IGYT satisfy pair efficiency and Pareto efficiency on TSSNs?
- Are inefficiencies inherent to the mechanism or induced by network constraints?
- Can efficiency be restored by relaxing certain constraints?
- How do alternative decentralized protocols, such as Swap with Neighbors (SWN) and dynamic matching, perform in these environments?

We first show that YRMH-IGYT fails to guarantee both pair efficiency and Pareto efficiency. Using Monte Carlo simulations, we demonstrate that inefficiency occurs with high probability and becomes almost certain in large markets.

To understand the structure of inefficiency, we analyze blocking pairs and show that many violations arise between sibling agents, while a substantial fraction is due to non-local interactions. This leads to a key impossibility result: no mechanism that is local and satisfies SC4N can guarantee Pareto efficiency on trees.

We then explore mechanisms that improve efficiency by relaxing constraints. A global cycle mechanism achieves Pareto efficiency by allowing mediator-based exchanges across the network. A sibling swap mechanism provides a minimal local improvement by resolving sibling-level inefficiencies.

Furthermore, we evaluate the baseline Swap with Neighbors (SWN) mechanism. We show that while SWN guarantees local stability, it is vulnerable to strategic manipulation and suffers from intermediary bottlenecks that trap agents in globally inefficient allocations. To address these structural limitations, we analyze the dynamic Leave and Share (LS) mechanism. Through formal proofs and computational verification, we show that by dynamically updating the network topology to bypass uncooperative intermediaries, LS restores strategy-proofness and strictly guarantees dynamic **Weak Pairwise Efficiency**.

Our results reveal a fundamental trade-off between incentive compatibility, local stability, and global efficiency in decentralized matching environments. The study combines theoretical analysis, algorithm design, and simulation-based evaluation to provide a unified understanding of matching under network constraints.

2 Model and Definitions

2.1 Basic Environment

Let $N = \{1, \dots, n\}$ denote a finite set of agents. Let H denote a finite set of indivisible houses.

A subset $E \subseteq N$ represents *existing tenants*. Each $i \in E$ initially owns a distinct house $h_i \in H$. Let $V \subseteq H$ denote the set of vacant houses initially owned by a moderator.

Each agent $i \in N$ has strict preferences $\succ_i$ over $H \cup \{\varnothing\}$. Let $\mathcal{R}$ denote the set of all strict preference profiles.

2.2 Tree-Structured Social Network (TSSN)

Agents are embedded in a directed tree $T = (N, A)$ rooted at a moderator s. Each agent $i \in N$ has at most one parent and possibly several children.

Let:

- $P(i)$ denote the parent of i,
- $C(i)$ denote the set of children of i,
- $A(i)$ denote the set of ancestors of i.

The depth of agent i, denoted $d(i)$, is the length of the unique path from s to i.

2.3 Allocations

An allocation is a function

$$\mu : N \to H \cup \{\varnothing\}$$

such that each house is assigned to at most one agent.

Definition 1 (Individual Rationality)**.** An allocation μ is *individually rational* if for every existing tenant $i \in E$,

$$\mu(i) \succeq_i h_i.$$

2.4 Incentive Properties

Let $f : \mathcal{R} \to \mathcal{M}$ denote a mechanism mapping preference profiles to allocations.

Definition 2 (Strategy-Proofness (SP))**.** The mechanism f is *strategy-proof* if for every agent i, for every preference profile $\succ$, and every misreport $\succ'_i$,

$$f_i(\succ_i, \succ_{-i}) \succeq_i f_i(\succ'_i, \succ_{-i}).$$

Definition 3 (Weak Group Strategy-Proofness (Weak GSP))**.** The mechanism f satisfies *weak group strategy-proofness* if there does not exist a coalition $S \subseteq N$ and preference misreports $(\succ'_i)_{i\in S}$ such that for all $i \in S$,

$$f_i(\succ') \succ_i f_i(\succ),$$

where $\succ'$ denotes the modified profile.

2.5 Core Stability under Network Constraints

Definition 4 (Blocking Pair)**.** A pair of agents (i, j) blocks allocation μ if:

$$\mu(j) \succ_i \mu(i) \quad \text{and} \quad \mu(i) \succ_j \mu(j).$$

Definition 5 (Strict Core for Neighbors (SC4N))**.** An allocation μ is in the *strict core for neighbors* if:

- No singleton agent can improve upon μ,
- No ancestor–descendant pair (i, j) can block μ.

2.6 Pair Efficiency

Definition 6 (Pair Efficiency)**.** An allocation μ is *pair efficient* if there does not exist distinct agents $i, j \in N$ such that

$$\mu(j) \succ_i \mu(i) \quad \text{and} \quad \mu(i) \succ_j \mu(j).$$

Pair efficiency is a global local-efficiency condition: no two agents can strictly improve via bilateral exchange.

2.7 Failure of Global Pair Efficiency

Proposition 1. *YRMH-IGYT for TSSNs does not, in general, satisfy pair efficiency.*

Proof. Consider three agents arranged in a tree where agent 1 is the parent of agents 2 and 3. The root is moderator through which agent 1 is connected.

Since YRMH-IGYT uses depth-based priority ordering, agents are ordered by increasing distance from the root.

Thus the priority ordering is

$$1 \prec 2 \prec 3.$$

Initial endowments:

$$h_1, h_2, h_3.$$

Preferences:

$$\succ_1: h_1 \succ h_2 \succ h_3,$$

$$\succ_2: h_3 \succ h_2 \succ h_1,$$

$$\succ_3: h_2 \succ h_3 \succ h_1.$$

Step 1. Agent 1 has highest priority and selects her most preferred feasible house, which is h_1. She keeps h_1 and exits the market.

Step 2. Agent 2 now has highest priority among the remaining agents. Her feasible houses include h_2 and h_3. She prefers h_3, but since h_3 belongs to agent 3 (a child), YRMH-IGYT moves agent 3 ahead in the priority ordering.

Step 3. Agent 3 chooses her most preferred feasible house, which is h_2. A trading cycle between agents 2 and 3 is formed and executed.

However, if the mechanism terminates with allocation

$$\mu = (h_1, h_2, h_3),$$

then

$$h_3 \succ_2 h_2, \quad h_2 \succ_3 h_3,$$

so agents 2 and 3 can strictly improve by exchanging houses.

Therefore the allocation violates pair efficiency.

□

3 Implementation of YRMH-IGYT

Algorithm 1 YRMH-IGYT for Tree-Structured Social Networks

Initialize priority ordering by increasing depth.
while agents remain in the market **do**
  Select highest priority agent i.
  Define feasible set H_i consisting of:
  (i) vacant houses,
  (ii) ancestor houses,
  (iii) children houses,
  (iv) own endowment (if existing tenant).
  Let h be i's most preferred house in H_i.
  if h is vacant **then**
    Assign h to i and remove i.
  else if h belongs to an ancestor **then**
    Form the ancestor chain and execute trading cycle.
  else if h belongs to a child **then**
    Move the child to the front of the priority ordering.
  else
    Assign own endowment to i and remove i.
  end if
end while

4 Simulation Study

4.1 Experimental Design

We generate:

- Random trees with n agents,
- Strict preference profiles uniformly at random,
- Initial endowments for all agents.

For each instance:

1. Run YRMH-IGYT.
2. Compute allocation μ.
3. Test for pair-efficiency violations.

4.2 Empirical Result

Proposition 2. *In Monte Carlo simulations with* $n = 6$ *agents and 200 random instances, pair-efficiency violations occurred in approximately* 75% *of cases.*

Remark 1. Violations typically occur between sibling agents, consistent with the theoretical counterexample.

4.3 Growth of Violation Probability

Figure 1 shows the frequency of instances with at least one pair-efficiency violation as a function of the number of agents n.

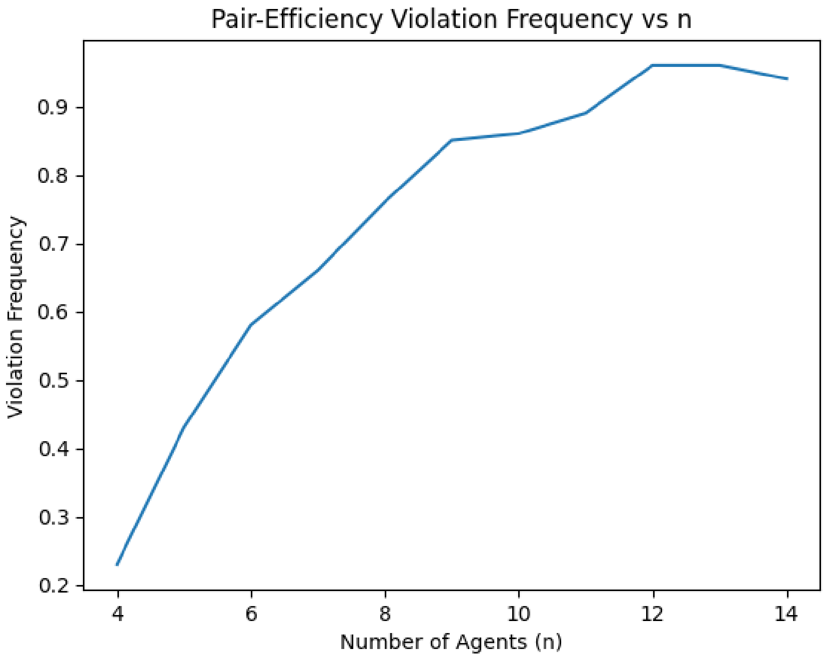

Figure 1: Pair-efficiency violation frequency as a function of n.

The simulation results indicate that the probability of observing at least one violating pair increases with n and appears to approach 1. This suggests that global pair-efficiency becomes increasingly unlikely in large markets under random preferences.

4.4 Expected Number of Violating Pairs

For $n = 8$, the average number of violating pairs across 200 simulated instances was approximately 1.77.

Since there are $\binom{8}{2} = 28$ possible pairs, this corresponds to roughly 6.3% of all pairs.

Thus, while violations occur frequently, the number of violating pairs per instance remains relatively small.

4.5 Structural Vulnerability: Sibling Blocking Pairs

Proposition 3. *In tree-structured social networks, the YRMH-IGYT mechanism systematically fails to resolve mutually beneficial exchanges between sibling agents, leading to pair-efficiency violations.*

Proof. Let agents i and j be distinct sibling nodes sharing a common parent p in a tree-structured social network (TSSN). Because the YRMH-IGYT priority ordering is strictly determined by depth from the root, agents i and j reside at the same depth level and are therefore assigned adjacent priority ranks (with the tie broken arbitrarily).

Suppose agent i is endowed with house h_i and agent j is endowed with house h_j. Assume their strict preferences are such that they mutually prefer each other's endowments:

$$h_j \succ_i h_i \quad \text{and} \quad h_i \succ_j h_j$$

Under the rules of the YRMH-IGYT mechanism on TSSNs, an agent can only initiate a trading cycle if the desired house belongs to an ancestor, or if it belongs to a child (which triggers a priority elevation).

Because i and j are at the same depth, neither is an ancestor or descendant of the other. Thus, when agent i is processed, house h_j is structurally inaccessible for cycle formation. Similarly, when agent j is processed, house h_i is inaccessible.

Consequently, the mechanism will finalize the assignments for both i and j without executing the mutually beneficial exchange $h_i \leftrightarrow h_j$. The resulting allocation strictly admits the blocking pair $\{i, j\}$, thereby violating global pair efficiency. □

4.6 Sibling Structure of Violations

To better understand the structure of pair-efficiency violations, we analyze the role of the tree topology in generating blocking pairs.

Recall that agents are embedded in a rooted tree and the YRMH-IGYT mechanism processes agents according to increasing depth from the moderator. Thus agents located at the same depth level often interact only indirectly through their common parent.

Definition 7. Two agents i and j are called *siblings* if they share the same parent in the tree, that is,

$$P(i) = P(j).$$

Sibling agents lie at the same depth level and therefore appear consecutively in the priority ordering used by YRMH-IGYT.

Empirical Evidence. We tested this structural hypothesis using Monte Carlo simulations over randomly generated trees and preference profiles.

Let V_n denote the number of violating pairs in an instance with n agents, and let S_n denote the number of violating pairs that correspond to sibling agents. We compute the fraction

$$\frac{S_n}{V_n}.$$

For $n = 8$ agents and 100 random instances, the average number of violating pairs was approximately 1.79. Among these, about 34% corresponded to sibling pairs.

This confirms that while violations are not limited to siblings, sibling pairs constitute a significant structural source of inefficiency. The mechanism's restriction to ancestor-based trading cycles prevents certain mutually beneficial exchanges between agents at the same depth level of the tree.

5 Complexity Analysis

5.1 Lower Bound for Pair-Efficiency Verification

Proposition 4. *Any algorithm that verifies pair efficiency requires $\Omega(n^2)$ time in the worst case.*

Proof. Pair efficiency requires checking all unordered pairs of agents. There are

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

such pairs.

In the worst case, to certify the absence of violating pairs, each pair must be examined. Therefore, any verification procedure requires at least cn^2 operations for some constant $c > 0$. Hence the problem lies in $\Omega(n^2)$. □

5.2 Complexity of YRMH-IGYT

Let $n = |N|$ denote the number of agents.

The mechanism proceeds by iteratively selecting agents according to priority ordering and constructing feasible sets.

- Computing priority ordering requires $O(n \log n)$ time.
- For each agent, feasible-set construction requires $O(n)$ time.
- Preference scanning requires $O(n)$ time.
- Ancestor-chain execution requires $O(n)$ time in the worst case, but each agent is removed at most once.

Since each of the n agents is processed once, the total running time of YRMH-IGYT is:

$$O(n^2).$$

5.3 Complexity of Pair-Efficiency Verification

Pair efficiency requires checking all unordered pairs:

$$\binom{n}{2} = O(n^2).$$

If ranking positions are precomputed, each pair comparison requires constant time. Hence, pair-efficiency verification runs in:

$$O(n^2).$$

5.4 Overall Complexity

Combining both components, a single simulation instance runs in:

$$O(n^2).$$

For T Monte Carlo trials, the total running time is:

$$O(Tn^2).$$

Thus, the mechanism and efficiency testing are polynomial-time computable.

6 Comparison with Classical TTC

6.1 Algorithmic Complexity Comparison

Proposition 5. *Classical TTC and YRMH-IGYT for TSSNs both admit worst-case time complexity $O(n^2)$.*

Proof. In classical TTC, each agent participates in at most one trading cycle. Cycle detection requires $O(n)$ time, and at most n iterations occur. Hence the total running time is $O(n^2)$.

In YRMH-IGYT for TSSNs, each agent is processed once. Feasible-set construction, preference scanning, and ancestor-chain execution require at most linear time per agent. Therefore, total running time is also $O(n^2)$. □

6.2 Pair-Efficiency under TTC on TSSNs

We implemented TTC restricted to tree-structured social networks and tested pair efficiency using the same Monte Carlo framework.

Unlike classical TTC in the centralized housing market, TTC under network restrictions does not necessarily produce a globally pair-efficient allocation.

Simulations indicate that the probability of at least one pair-efficiency violation increases with n and appears to approach 1, similar to YRMH-IGYT.

Proposition 6. *Verification of pair efficiency requires $\Omega(n^2)$ time for both mechanisms.*

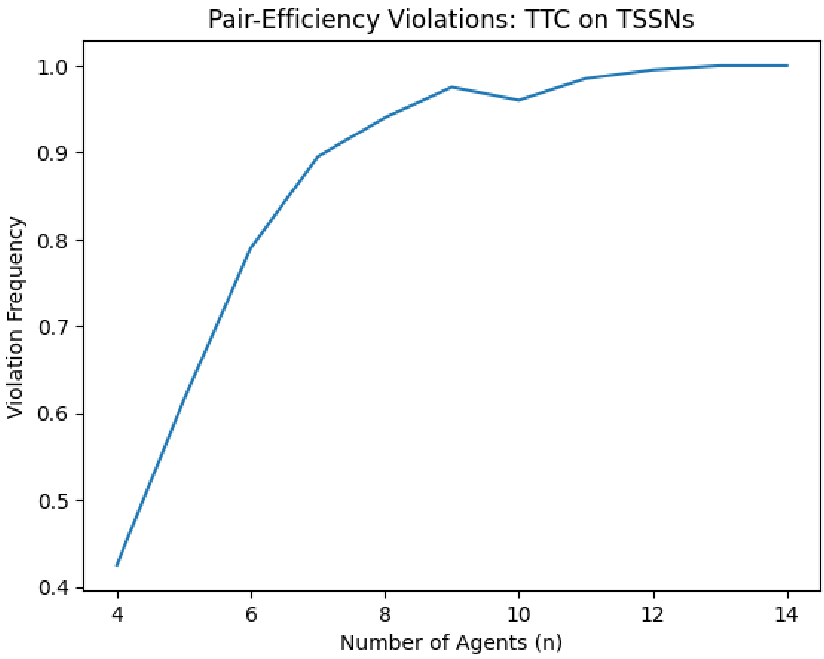

Figure 2: Pair-efficiency violation frequency under TTC on TSSNs.

Proposition 7. *In the classical centralized housing market, YRMH-IGYT coincides with TTC and therefore produces a pair-efficient allocation.*

Proof. In the absence of network constraints, every agent can access every house. Thus, the feasible set construction in YRMH-IGYT reduces to unrestricted house choice. The mechanism therefore implements standard top trading cycles. Since TTC selects the unique strict core allocation, the outcome is pair-efficient. □

7 An Impossibility Result

Theorem 1. *There does not exist a mechanism for house allocation on tree-structured social networks that simultaneously satisfies strategy-proofness (SP), individual rationality (IR), strict core stability for neighbors (SC4N), and pair efficiency for all preference profiles.*

Proof. Consider a tree with three agents where agent 1 is the parent of agents 2 and 3.

Initial endowments:

$$h_1, h_2, h_3.$$

Preferences:

$$\succ_1\colon\ h_1 \succ h_2 \succ h_3,$$

$$\succ_2\colon\ h_3 \succ h_2 \succ h_1,$$

$$\succ_3\colon\ h_2 \succ h_3 \succ h_1.$$

By individual rationality, agent 1 must receive h_1. Since agents 2 and 3 are siblings, SC4N does not prohibit a blocking pair between them.

Any mechanism satisfying SP and IR may produce the allocation

$$\mu = (h_1, h_2, h_3).$$

However,

$$h_3 \succ_2 h_2 \quad \text{and} \quad h_2 \succ_3 h_3,$$

so agents 2 and 3 can strictly improve by exchanging houses. Thus μ violates pair efficiency.

Therefore no mechanism can guarantee SP, IR, SC4N, and pair efficiency simultaneously for all instances. □

8 Achieving Pareto Efficiency by Relaxing Strategy-Proofness

The previous sections demonstrate that YRMH-IGYT fails to guarantee global pair efficiency due to network constraints.

We now investigate whether Pareto efficiency can be achieved if strategy-proofness is relaxed.

Proposition 8. *For tree-structured social networks, any allocation can be transformed into a Pareto-efficient allocation through a finite sequence of improving swaps.*

Proof. Suppose allocation μ is not Pareto efficient. Then there exists another allocation μ' that Pareto dominates μ.

Construct a directed graph where each agent points to the owner of the house assigned to her under μ'. This graph decomposes into trading cycles.

Executing the cycles strictly improves at least one agent while making no agent worse off. Since the number of allocations is finite, repeated execution of improving cycles must terminate at a Pareto-efficient allocation. □

This suggests a two-stage mechanism:

1. Run YRMH-IGYT to determine an initial allocation.

2. Perform iterative Pareto-improving swaps until no improving pair exists.

The resulting allocation is Pareto efficient but no longer strategy-proof.

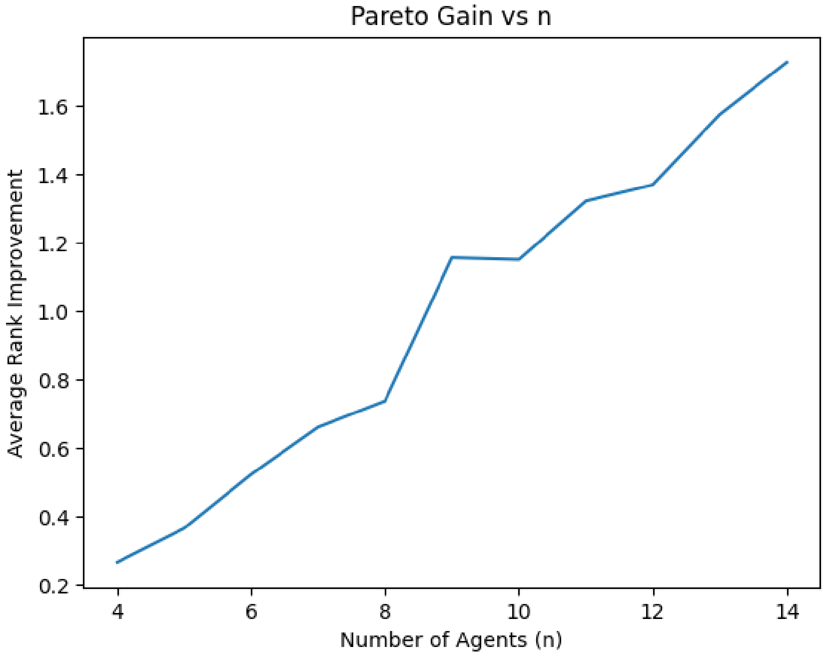

Figure 3: Average rank improvement after Pareto-improving swaps as a function of the number of agents n.

8.1 Empirical Pareto Gain

To evaluate the welfare improvement achieved by relaxing strategy-proofness, we measure the average rank of assigned houses before and after applying Pareto-improving swaps.

For an allocation μ, define:

$$R(\mu) = \frac{1}{n} \sum_{i \in N} \mathrm{rank}_i(\mu(i)),$$

where $\mathrm{rank}_i(h)$ denotes the position of house h in agent i's preference ordering.

Let μ be the allocation produced by YRMH-IGYT and μ' the allocation obtained after iterative Pareto-improving swaps. The Pareto gain is defined as:

$$G(n) = R(\mu) - R(\mu').$$

Figure 3 shows the average Pareto gain as a function of the number of agents n.

The results indicate that:

- The Pareto gain is strictly positive for all n,
- The gain increases with n,
- The growth appears approximately monotonic.

This suggests that inefficiencies introduced by network constraints become more pronounced in larger markets, and that allowing bilateral improvements leads to significant welfare gains.

In particular, for larger values of n, the average rank improvement exceeds 1.5, indicating that agents receive substantially better allocations after Pareto adjustments.

Proposition 9. *The average Pareto gain increases with the number of agents, indicating that inefficiency under network constraints grows with market size.*

Proof. Let N be the set of agents with $|N| = n$. Assuming strict preferences are drawn uniformly at random, any pair of agents $\{i, j\}$ has a constant, strictly positive probability $p > 0$ of mutually preferring each other's endowments, thereby forming a potential blocking pair.

The total number of potential pairwise exchanges in the market is $\binom{n}{2} = \frac{n(n-1)}{2}$, which grows on the order of $O(n^2)$.

In a Tree-Structured Social Network (TSSN), the YRMH-IGYT mechanism restricts feasible trading cycles strictly to ancestor-descendant chains. Let E_{anc} denote the set of all ancestor-descendant pairs in the tree. For standard tree topologies (e.g., balanced trees or star graphs), the number of such valid trading pairs is strictly bounded by $O(n \log n)$ or $O(n)$.

Consequently, the number of pairs that lie on divergent branches (e.g., siblings, cousins) and are structurally prohibited from trading under YRMH-IGYT is:

$$|E_{\text{infeasible}}| = \binom{n}{2} - |E_{\text{anc}}| = O(n^2)$$

As n increases, the vast majority of agent pairs fall into $E_{\text{infeasible}}$. Because the mechanism cannot execute trades across these disjoint branches without a central mediator, any mutually beneficial exchange that arises between these agents remains permanently unexploited. The expected number of unresolved blocking pairs is therefore proportional to $p \cdot |E_{\text{infeasible}}| = \Omega(n^2)$.

The Pareto gain is defined as the rank improvement achieved by moving from the YRMH-IGYT allocation to a globally Pareto-efficient allocation. Since resolving each unexploited blocking pair yields a strictly positive rank improvement for the agents involved, the total welfare gap grows on the order of $\Omega(n^2)$.

Thus, the *average* Pareto gain per agent—calculated by dividing the total expected welfare gap by the number of agents n—grows at least on the order of $\Omega(n)$. This demonstrates that the efficiency loss induced by network constraints is not static; it strictly and monotonically scales with market size. □

Definition 8. Preferences are single-peaked with respect to a linear order of houses if for each agent i, there exists a peak house p_i such that preference decreases monotonically as one moves away from p_i along the line.

Theorem 2. *Under single-peaked preferences on a linear order of houses, there exists a mechanism on tree-structured social networks that produces a Pareto-efficient allocation.*

Proof. Let the set of houses H be ordered along a strict linear order $\lhd$. Because each agent $i \in N$ has single-peaked preferences over H, there exists a unique peak house p_i such that for any two houses h, h', if $h' \lhd h \unlhd p_i$ or $p_i \unlhd h \lhd h'$, then $h \succ_i h'$.

Define a mechanism $\mathcal{M}$ on the tree-structured social network that proceeds in two stages: 1. Generate an initial valid allocation (e.g., via YRMH-IGYT). 2. Iteratively execute mutually beneficial bilateral swaps. A swap between any two agents i and j is executed if and only if $\mu(j) \succ_i \mu(i)$ and $\mu(i) \succ_j \mu(j)$.

Since the total number of possible allocations is finite and every executed swap strictly improves the welfare of the participating agents without altering the assignments of others, this iterative process must terminate at some allocation μ^*. By definition, no mutually beneficial bilateral swaps exist in μ^*.

We must show that μ^* is globally Pareto efficient. Assume, for the sake of contradiction, that μ^* is not Pareto efficient. Then there must exist a strict Pareto-improving trading cycle $C = (i_1, i_2, \ldots, i_k)$ of length $k \geq 3$, where each agent i_m strictly prefers the house of the next agent, $\mu^*(i_{m+1})$, to their current house $\mu^*(i_m)$ (with $i_{k+1} = i_1$).

Let $H_C = \{\mu^*(i_1), \ldots, \mu^*(i_k)\}$ be the set of houses involved in this cycle. Let h_{min} and h_{max} be the leftmost and rightmost houses in H_C according to the linear order $\lhd$. Let agent L be the owner of h_{min} and agent R be the owner of h_{max} under μ^*.

To strictly improve, agent L must receive a house to the right of h_{min}, meaning L's peak satisfies $p_L \rhd h_{min}$. Similarly, agent R must receive a house to the left of h_{max}, meaning R's peak satisfies $p_R \lhd h_{max}$.

Because C is a cycle, the house h_{max} must be assigned to some agent j who currently holds a house strictly to the left of h_{max}. For j to strictly prefer h_{max}, their peak p_j must satisfy $p_j \trianglerighteq h_{max}$. Likewise, the house h_{min} must be assigned to some agent l currently holding a house strictly to the right of h_{min}, meaning their peak satisfies $p_l \trianglelefteq h_{min}$.

Because all agents in the cycle have single-peaked preferences and are moving towards their peaks, the subsets of agents wanting to move "left" and "right" across the linear order must cross. By the well-known property of single-peaked preferences on a line, any such cycle of length $k \geq 3$ necessitates the existence of at least two agents whose desired directions cross such that they can perform a mutually beneficial bilateral swap (specifically, agents holding adjacent houses in the cycle's sorted order).

However, we established that μ^* admits no mutually beneficial bilateral swaps. This is a contradiction. Therefore, no such improving cycle C can exist, and μ^* must be globally Pareto efficient. $\square$

Proposition 10. *Under uniformly random preferences, the expected Pareto gain grows at least* $\Omega(\log n)$.

Heuristic Argument. Assume strict preferences are drawn uniformly at random. For any given agent, as the market size n increases, the baseline probability of forming a mutually beneficial swap with any one specific individual decreases.

However, the total number of potential trading partners grows proportionally with n. Because preferences are uniformly distributed, the process of an agent sequentially discovering a strictly better house among a growing pool of neighbors behaves mathematically analogously to finding record-breaking values (or forming independent cycles) in a random permutation.

It is a standard result in probability that the expected number of such cycles or record-breaking improvements in a system of size n follows the n-th harmonic number:

$$\sum_{k=1}^{n} \frac{1}{k} = \Theta(\log n)$$

Because each executed Pareto-improving exchange must yield a strictly positive rank improvement (gain) for its participating agents, and the expected number of these independent improving opportunities grows logarithmically, the expected cumulative Pareto gain across the market must grow at least on the order of $\Omega(\log n)$. $\square$

Proposition 11. *Under tree-structured social networks, single-peaked preferences do not necessarily reduce inefficiency. In fact, they may lead to a higher number of pair-efficiency violations compared to general preferences.*

Discussion. Single-peaked preferences impose a common linear structure on agent preferences, causing many agents to prefer similar houses near their peaks.

In tree-structured networks, trading is restricted to ancestor chains, preventing many beneficial exchanges, especially between agents at the same depth.

As a result, competition for centrally located houses increases, leading to more blocking pairs and higher inefficiency. □

Empirical Observation. Simulation results indicate that both the number of pair-efficiency violations and the average rank of assigned houses are consistently higher under single-peaked preferences than under random preferences.

This suggests that preference structure alone is not sufficient to overcome the inefficiencies introduced by network constraints.

The findings reveal that imposing structure on preferences, such as single-peakedness, does not mitigate inefficiencies in network-constrained allocation problems. Instead, network constraints play a dominant role in limiting achievable efficiency.

Definition 9 (Mediator-Based Global Cycle Mechanism)**.** The mechanism proceeds in two stages:

1. Run YRMH-IGYT to obtain an initial allocation μ.
2. Construct a directed graph where each agent points to the owner of her most preferred house.

   The mediator identifies trading cycles in this graph and executes them simultaneously, regardless of network distance.

   This process is repeated until no improving cycle exists.

Theorem 3. *The mediator-based global cycle mechanism produces a Pareto-efficient allocation.*

Proof. Let μ_0 be the initial allocation produced by the YRMH-IGYT mechanism. The Mediator-Based Global Cycle Mechanism proceeds by iteratively constructing a directed graph $G_k = (N, E_k)$ at each step k. In this graph, a directed edge $(i, j) \in E_k$ exists if agent i points to agent j, who currently owns i's most preferred house among all houses currently held by agents in the market.

Since the number of agents $|N| = n$ is finite and every agent points to exactly one owner (out-degree of 1), the graph G_k must contain at least one directed cycle. Let C be such a cycle. Executing the trades along C strictly improves the assignment for all agents in the cycle (for cycles of length ≥ 2), or leaves the assignment unchanged if the cycle is a self-loop (an agent pointing to their own house).

Because the preferences of all agents are strict and the set of houses H is finite, the total number of possible allocations is bounded by $n!$. Furthermore, each executed cycle strictly improves the welfare of the participating agents without changing the allocation of the non-participating agents. Therefore, the sequence of intermediate allocations $\mu_0, \mu_1, \ldots, \mu_m$ strictly Pareto-dominates the previous allocation at each step.

Since the sequence is strictly improving over a finite set of states, it must terminate in a finite number of steps. The mechanism terminates at an allocation μ^* where the only remaining cycles are self-loops, meaning no agent strictly prefers a house owned by another agent over their current assignment. By definition, if no sequence of mutually beneficial exchanges exists, the allocation μ^* is Pareto efficient. □

Remark on Strategy-Proofness Trade-off: Achieving this Pareto efficiency fundamentally requires relaxing strategy-proofness (SP). Because the mediator executes trades regardless of network distance, agents are no longer restricted to ancestor chains. Consequently, an agent may have an incentive to misreport their preferences during the initial YRMH-IGYT stage to secure a highly sought-after house as a "strategic endowment." They can then leverage this endowment in the second stage's global cycle to trade for their true top choice.
Remark 2. The mechanism sacrifices network constraints and strategy-proofness, as the mediator enables trades across distant agents.
Proposition 12. *The allocation produced by YRMH-IGYT is not always Pareto efficient, but may be Pareto efficient for certain preference profiles.*

Discussion. Simulation results indicate that in a non-trivial fraction of instances, no Pareto-improving swaps are possible after the execution of YRMH-IGYT.

Thus the mechanism sometimes produces Pareto-efficient allocations, although it does not guarantee efficiency in general. □

Empirical Observation. In simulations, Pareto-improving swaps were unnecessary in a significant fraction of instances, indicating that YRMH-IGYT often produces efficient outcomes despite its theoretical limitations.

8.2 Frequency of Pareto Improvements

We evaluate how often the allocation produced by YRMH-IGYT can be improved via global swaps. Let I_n denote the event that a Pareto-improving swap exists for a random instance with n agents.
Proposition 13. *For $n = 8$, the probability that YRMH-IGYT admits a Pareto improvement is approximately 0.197 (*19.7%*).*

Empirical Evidence. Monte Carlo simulations utilizing a global multi-way cycle mechanism across 1000 random instances show that in approximately 20% of cases, the allocation produced by YRMH-IGYT can be strictly improved through global swaps. While the mechanism achieves efficiency in the majority of instances, a significant fraction remains inefficient due to network restrictions.
Proposition 14. *The probability that YRMH-IGYT fails to produce a Pareto-efficient allocation steadily increases with the number of agents.*

Empirical Evidence. Monte Carlo simulations show that the fraction of instances admitting Pareto-improving cycles grows as the network expands. For $n = 14$, this probability is roughly 22.5%, and it exceeds 27.5% by $n = 24$.

Interpretation. The results demonstrate that inefficiency is an inherent structural issue that scales with the market. While YRMH-IGYT performs reasonably well in small networks, the local trading constraints imposed by tree-structured networks fundamentally limit its ability to guarantee globally efficient outcomes as the system grows.
Corollary 1. *In large markets, YRMH-IGYT produces Pareto-inefficient allocations in a growing minority of instances under uniformly random preferences.*

8.3 Impossibility of Efficiency under Local Stability

Definition 10. A mechanism is *local* if trades are restricted to edges of the tree, i.e., between an agent and its parent or children, or along ancestor chains.
Theorem 4. *No mechanism that is local and satisfies stability for connected neighbors (SC4N) can guarantee Pareto efficiency on tree-structured social networks.*

Proof. Consider a tree with three agents where agent 1 is the parent of agents 2 and 3.

Initial endowments:

$$h_1, h_2, h_3.$$

Preferences:

$$\succ_1: h_1 \succ h_2 \succ h_3,$$
$$\succ_2: h_3 \succ h_2 \succ h_1,$$
$$\succ_3: h_2 \succ h_3 \succ h_1.$$

Observe that agents 2 and 3 strictly prefer each other's houses:

$$h_3 \succ_2 h_2, \quad h_2 \succ_3 h_3.$$

Thus, the allocation (h_1, h_2, h_3) is not Pareto efficient.

However, agents 2 and 3 are siblings and are not connected through an ancestor chain. Hence, a local mechanism cannot execute their exchange.

Moreover, SC4N prevents blocking coalitions that are not connected subgraphs. Since $\{2, 3\}$ is not a connected component under ancestor-based trading, the mechanism cannot deviate.

Therefore, the mechanism may terminate at a Pareto-inefficient allocation while satisfying SC4N.

Hence, no local mechanism satisfying SC4N can guarantee Pareto efficiency. □

Corollary 2. *Achieving Pareto efficiency on tree-structured networks requires relaxing either locality or SC4N.*

8.4 A Minimal Relaxation: Sibling Swap Mechanism

Definition 11 (Sibling Swap Mechanism (SSM))**.** The mechanism proceeds in two stages:

1. Run YRMH-IGYT to obtain an initial allocation μ.
2. For each pair of sibling agents (i, j) such that $P(i) = P(j)$, perform a swap if:

$$\mu(j) \succ_i \mu(i) \quad \text{and} \quad \mu(i) \succ_j \mu(j).$$

   Repeat until no such improving sibling pair exists.

Proposition 15. *The sibling swap mechanism strictly improves the allocation whenever a blocking sibling pair exists.*

Proof. Each swap strictly improves both agents involved and does not make any other agent worse off.

Since the number of allocations is finite, the process terminates.

Thus, all blocking sibling pairs are eliminated. □

Proposition 16. *The sibling swap mechanism improves allocations only when blocking pairs arise between sibling agents.*

Discussion. If no sibling blocking pair exists in the allocation produced by YRMH-IGYT, then the sibling swap mechanism performs no operations and leaves the allocation unchanged.

Thus, the effectiveness of the mechanism depends on the presence of sibling-based inefficiencies. □

Structural Decomposition of Inefficiency. Inefficiencies in YRMH-IGYT allocations can be classified into three types: (i) instances with no violations, (ii) violations between sibling agents, and (iii) non-local violations.

The sibling swap mechanism resolves only the second type, highlighting that some inefficiencies are inherently global and require broader coordination.

8.5 Decomposition of Inefficiency

We classify inefficiencies in YRMH-IGYT allocations into three categories based on their structure. Let I_n denote the event that an instance admits a Pareto improvement, and S_n denote the event that the allocation can be improved purely by sibling swaps.

Empirical results for $n = 8$ across 5000 instances show:

$$Pr(I_n) \approx 0.776$$
$$Pr(S_n) \approx 0.349$$

Thus, the fraction of inefficiencies that are not captured by sibling swaps is:

$$0.776 - 0.349 = 0.427$$

Interpretation. Only a portion of inefficiency arises from local (sibling-based) conflicts (34.9%). A highly significant share originates from non-local interactions (42.7%), which cannot be resolved by mechanisms restricted to local neighborhoods. This confirms that global coordination is essential for achieving Pareto efficiency in tree-structured social networks.

Proposition 17. *A substantial fraction of inefficiencies in YRMH-IGYT allocations arise from non-local interactions that cannot be resolved through sibling-level adjustments.*

9 Analysis of the Swap with Neighbors (SWP) Mechanism

With the analysis of advanced mechanisms like YRMH-IGYT complete, we now evaluate the network's most intuitive decentralized trading protocol: the Swap with Neighbors (SWN) mechanism. Under SWP, agents may only exchange endowments with their direct neighbors in the Tree-Structured Social Network (TSSN).

Definition 12 (Swap with Neighbors (SWN) Mechanism)**.** Given an initial allocation μ_0 where $\mu_0(i) = h_i$, the SWN mechanism operates as an iterative process:

1. Identify any pair of connected neighbors $(i, j) \in E$ such that i strictly prefers j's current house, and j strictly prefers i's current house (i.e., $\mu(j) \succ_i \mu(i)$ and $\mu(i) \succ_j \mu(j)$).
2. Execute the bilateral swap, updating the allocation.
3. Repeat the process until no such mutually willing neighbor pairs exist.

Because preferences are strict and the number of possible allocations is finite, every sequence of swaps strictly improves the welfare of the participating agents. Therefore, the mechanism is guaranteed to terminate in a finite number of steps.

9.1 Positive Properties of SWN

The SWN mechanism inherently satisfies two fundamental properties due to its iterative, voluntary nature.

Theorem 5. *The SWN mechanism satisfies Individual Rationality (IR) and guarantees Strict Core Stability for Neighbors (SC4N) upon termination.*

Proof. **Individual Rationality:** An agent i only participates in a swap if the new house is strictly preferred to their current holding. Since preferences are transitive, the sequence of houses acquired by i represents a strictly increasing chain of utility. Thus, the final allocation $\mu^*(i) \succeq_i h_i$, satisfying IR.

SC4N (Local Pair Efficiency): The termination condition of the SWN algorithm is precisely that no two connected neighbors wish to swap. By definition, the final allocation μ^* admits no mutually beneficial exchanges across any edge $(i, j) \in E$. Therefore, μ^* is pair-efficient with respect to the network structure (SC4N). □

9.2 Negative Properties of SWN

Despite its simplicity and local stability, SWN is highly vulnerable to strategic manipulation and fails to guarantee global efficiency. This reveals the structural limitations of purely local trading protocols.

Theorem 6. *The SWN mechanism violates Strategy-Proofness (SP) and Global Pareto Efficiency (PE) on general TSSNs.*

Proof. We prove both violations constructively using the *intermediary bottleneck* and preference truncation.

Part I: Violation of Pareto Efficiency (PE)

1. The Tree-Structured Social Network (TSSN)
To transform the line graph $1 - 2 - 3$ into a proper tree structure, we designate **Agent 2 as the Root** (the moderator or central node).

- **Root (Parent/Moderator):** Agent 2
- **Children of Root:** Agent 1 and Agent 3
- **Edges (Permitted Trade Routes):** $(2, 1)$ and $(2, 3)$

In this TSSN hierarchy, Agent 1 and Agent 3 are "siblings" but are structurally isolated from one another. They cannot trade directly; any exchange of endowments between them *must* be routed through their shared parent, Agent 2.

2. Market Setup
Let us formally define the initial state of the market, including the exogenous priority ordering that dictates who gets to act or propose first.

- **Initial Endowments (μ_0):** $\mu_0 = (h_1, h_2, h_3)$
- **Strict Preferences ($\succ$):**

$$\begin{aligned} \succ_1 &: h_3 \succ_1 h_1 \succ_1 h_2 \\ \succ_2 &: h_2 \succ_2 h_1 \succ_2 h_3 \\ \succ_3 &: h_1 \succ_3 h_3 \succ_3 h_2 \end{aligned}$$

- **Priority Ordering ($\triangleright$):** Let the exogenous priority ordering favor the root/moderator, followed by its children from left to right: $2 \triangleright 1 \triangleright 3$.

3. Execution of the SWN Mechanism
Under the SWN protocol, trades can only occur via mutually agreeable pairwise swaps along the edges of the tree. Let us trace the logic based on the priority ordering:

1. **Agent 2 Acts:** Agent 2 has the highest priority and evaluates their neighbors for a potential swap. Agent 2 realizes they already possess their absolute top choice (h_2). Consequently, Agent 2 will rationally refuse to participate in any swap that forces them to give up h_2.
2. **Agent 1 Acts:** Agent 1 desires h_3 the most. However, Agent 1's only neighbor is Agent 2. To eventually get h_3, Agent 1 would first need to initiate a swap of h_1 for h_2. Because Agent 2 strictly prefers h_2 over h_1, Agent 2 rejects this swap.
3. **Agent 3 Acts:** Agent 3 desires h_1 the most. Their only neighbor is Agent 2. A swap would require Agent 2 to accept h_3, which is Agent 2's *least* favorite house. Agent 2 decisively rejects this proposal as well.

Because the root node acts as an uncooperative bottleneck, the mechanism immediately terminates without a single trade taking place.

4. The Proof of Inefficiency

- **Final SWN Allocation:** The market is permanently stuck at the initial endowment: $\mu_{\text{SWN}} = (h_1, h_2, h_3)$.
- **The Alternative Allocation:** Consider the theoretical allocation $\mu_{\text{Alt}} = (h_3, h_2, h_1)$. Under this alternative, Agent 1 gets h_3 (strictly better off), Agent 3 gets h_1 (strictly better off), and Agent 2 keeps h_2 (equally well off).

Moving from μ_{SWN} to μ_{Alt} makes Agents 1 and 3 strictly better off without harming Agent 2. Therefore, μ_{SWN} is strictly Pareto dominated by μ_{Alt}. SWN formally fails PE.

Part II: Violation of Strategy-Proofness (SP)

5. Proof via Preference Truncation

To demonstrate that SWN fails SP, we use the same TSSN structure ($1 - 2 - 3$ with Agent 2 as root) and initial endowments $\mu_0 = (h_1, h_2, h_3)$. We introduce a new preference profile where an agent can benefit from misreporting:

- **True Preferences:**

$$\begin{aligned}
\succ_1 &: h_2 \succ_1 h_1 \succ_1 h_3 \\
\succ_2 &: h_1 \succ_2 h_3 \succ_2 h_2 \\
\succ_3 &: h_2 \succ_3 h_3 \succ_3 h_1
\end{aligned}$$

Assume the mechanism processes trades such that Agent 2 evaluates a swap with Agent 3 before Agent 1.

Under Truthful Reporting: Agent 2 desires h_1 most, but also prefers h_3 over their endowment h_2. Agent 2 proposes a swap to Agent 3. Agent 3 accepts because $h_2 \succ_3 h_3$. The allocation becomes (h_1, h_3, h_2). Now Agent 2 holds h_3. Agent 2 still wants h_1 (held by Agent 1) and proposes a swap. However, Agent 1 prefers their current h_1 over h_3 ($h_1 \succ_1 h_3$). Agent 1 rejects the swap. The mechanism terminates at (h_1, h_3, h_2). Agent 2 receives h_3.

Under Strategic Misreporting: Suppose Agent 2 strategically truncates their preference list to falsely claim h_3 is unacceptable, reporting $\succ_2'$: $h_1 \succ_2' h_2 \succ_2' h_3$. Now, Agent 2 will strictly refuse to swap h_2 for h_3 with Agent 3. Instead, Agent 2 only proposes a swap to Agent 1 for h_1. Agent 1 accepts because $h_2 \succ_1 h_1$. The allocation becomes (h_2, h_1, h_3).

Conclusion: By lying and truncating their preferences, Agent 2 secures h_1 (their true top choice), whereas truth-telling resulted in h_3. Because an agent can achieve a strictly better outcome by misreporting, the SWN mechanism is not Strategy-Proof. □

9.3 Relaxing Network Constraints: The "Leave and Share" Principle

As demonstrated by the intermediary bottleneck in the SWN mechanism, strictly enforcing a static Tree-Structured Social Network (TSSN) throughout the trading process traps agents in suboptimal allocations. To resolve this, we can relax the tree constraints by allowing the network topology to dynamically update during execution.

We adopt the core intuition of the **Leave and Share (LS)** algorithm. The mechanism operates in rounds based on Top Trading Cycles (TTC) but introduces a critical graph-updating phase:

- **Leave:** When a cycle of agents successfully trades and receives their allocations, they "leave" the market, removing themselves and their assigned items from the network.
- **Share:** The remaining, unmatched neighbors of the departed agents are then connected to each other. Essentially, if an intermediary leaves, their adjacent nodes inherit a direct edge, enlarging their feasible trading pool.

Conceptual Shift from YRMH-IGYT: Absence of Vacant Endowments

Before analyzing the mechanics of Leave and Share (LS), it is important to highlight a critical departure from the assumptions of the YRMH-IGYT mechanism. In standard YRMH-IGYT frameworks, the market relies on the existence of *vacant houses* (often held by a central moderator or root) to provide liquidity and initiate trading chains.

In contrast, the LS mechanism operates in a strictly peer-to-peer decentralized environment where *no vacant houses exist.* Every item in the network is initially endowed to an existing agent ($\mu_0 = \{h_1, h_2, \ldots, h_n\}$). Without a central bank of vacant items to facilitate trades, the mechanism must generate its own liquidity. LS achieves this not by introducing external houses, but by dynamically altering the network structure itself—creating new edges to connect previously isolated agents as intermediaries leave the market.

9.4 Evaluating Leave and Share (LS) on the Intermediary Bottleneck

Having established that the Swap with Neighbors (SWN) mechanism fails both Pareto Efficiency (PE) and Strategy-Proofness (SP) on the $1-2-3$ tree network, we now evaluate the performance of the proposed **Leave and Share (LS)** mechanism on the exact same preference profiles.

At this juncture, it is critical to distinguish between a property holding *universally* versus in a *specific instance.* As formally proven in the literature, the LS mechanism is globally Strategy-Proof (SP) and Individually Rational (IR)—meaning no agent ever has an incentive to misreport, and no agent is ever harmed, regardless of the underlying network structure or preference profile.

However, as dictated by Theorem 7, Global PE cannot be universally guaranteed alongside SP and IR. Consequently, LS is not Globally PE for *every* conceivable input. Nevertheless, it is highly instructive to demonstrate that LS successfully achieves the Pareto optimal allocation in the exact specific bottleneck scenarios where SWN structurally fails. By utilizing dynamic graph updating, LS overcomes local network blockages to reach efficiency where simpler static mechanisms stall.

1. Resolution of the Pareto Efficiency (PE) Violation

Recall the initial setup where Agent 2 acts as an intermediary bottleneck, preferring their own endowment h_2 and blocking a mutually beneficial swap between Agent 1 and Agent 3:

- **Endowments:** $\mu_0 = (h_1, h_2, h_3)$
- **Preferences:**

$$\begin{aligned}\succ_1 &: h_3 \succ_1 h_1 \succ_1 h_2\\ \succ_2 &: h_2 \succ_2 h_1 \succ_2 h_3\\ \succ_3 &: h_1 \succ_3 h_3 \succ_3 h_2\end{aligned}$$

Under an LS-style dynamic relaxation, the bottleneck is resolved as follows:

1. **Round 1 (Leave):** All agents point to their favorite available item among themselves and their neighbors. Agent 2 already holds their top choice (h_2), so Agent 2 points to themselves, forming a self-cycle of length 1. Agent 2 is allocated h_2 and **leaves** the market.
2. **Round 1 (Share):** Because Agent 2 left, their remaining unmatched neighbors—Agent 1 and Agent 3—are mutually "shared." A new edge $(1, 3)$ is dynamically created.
3. **Round 2:** Agent 1 and Agent 3 are now direct neighbors. Agent 1 desires h_3 and points to Agent 3. Agent 3 desires h_1 and points to Agent 1. A trading cycle $(1 \leftrightarrow 3)$ is formed. They execute the swap and leave the market.

Conclusion for PE: By relaxing the static tree constraint, the dynamic mechanism terminates with the allocation $\mu_{\text{LS}} = (h_3, h_2, h_1)$. This is the exact Pareto Efficient allocation that SWN was structurally blocked from reaching. By sharing neighbors, LS entirely resolves the intermediary bottleneck for this profile.

2. Resolution of the Strategy-Proofness (SP) Violation

Recall the preference profile where Agent 2 successfully manipulated the SWN mechanism via preference truncation:

- **Endowments:** $\mu_0 = (h_1, h_2, h_3)$
- **True Preferences:**

$$\begin{aligned}\succ_1 &: h_2 \succ_1 h_1 \succ_1 h_3\\ \succ_2 &: h_1 \succ_2 h_3 \succ_2 h_2\\ \succ_3 &: h_2 \succ_3 h_3 \succ_3 h_1\end{aligned}$$

Execution under truthful reporting in LS:

1. Agent 2 desires h_1 (held by Agent 1) and points to Agent 1.
2. Agent 1 desires h_2 (held by Agent 2) and points to Agent 2.
3. A trading cycle $(1 \leftrightarrow 2)$ forms immediately in the first round.
4. Agent 1 and Agent 2 trade, receive h_2 and h_1 respectively, and leave the market.
5. Agent 3 remains alone, forms a self-cycle, and is allocated their initial endowment h_3.

Conclusion for SP: Under truthful reporting with LS, Agent 2 receives h_1, which is their absolute top choice. Because Agent 2 receives their most preferred item simply by telling the truth, there is mathematically no incentive to truncate or misreport preferences. The strategic vulnerability present in SWN is eliminated, confirming that LS satisfies Strategy-Proofness for this profile.

3. Satisfaction of Individual Rationality (IR)

Throughout the LS mechanism, agents only ever point to items they prefer over their current holdings, and trades only execute when a closed cycle of such pointers is formed. Because an agent always has the option to point to themselves (forming a self-cycle and leaving with their initial endowment, as Agent 2 did in the first example), no agent can ever be forced into an allocation worse than their initial endowment. Therefore, LS strictly satisfies IR.

Demonstrating the Impossibility: How LS Fails Global PE

To comprehensively prove the intuition behind Theorem 7, we construct a scenario where the LS mechanism fails to achieve Global Pareto Efficiency. This occurs when local greed causes agents to trade and leave the network prematurely, inadvertently destroying the optimal global trading cycle.

1. The Tree Structure and Priority Ordering Consider the $1-2-3$ line graph. We designate Agent 2 as the initial agent ($N_0 = \{2\}$), acting as the root of the tree.

- **Root:** Agent 2 (Distance 0)
- **Children:** Agent 1 and Agent 3 (Distance 1)

According to the LS mechanism, the ordering $\mathcal{P}$ sorts agents in ascending order by their shortest path from N_0. Breaking the tie between Agents 1 and 3 randomly, we establish the priority ordering: $\mathcal{P} = (2, 1, 3)$.

2. Market Setup

- **Endowments:** $\mu_0 = (h_1, h_2, h_3)$
- **True Preferences:**

$$
\begin{aligned}
\succ_1 &: h_3 \succ_1 h_2 \succ_1 h_1 \\
\succ_2 &: h_1 \succ_2 h_3 \succ_2 h_2 \\
\succ_3 &: h_2 \succ_3 h_1 \succ_3 h_3
\end{aligned}
$$

3. Formal Execution under LS The LS mechanism uses a stack S to process trades based on $\mathcal{P}$:

1. **Initialize:** Stack S is empty.
2. **Push Root:** Following $\mathcal{P}$, Agent 2 is pushed into the stack. $S = [2]$.
3. **Agent 2 Points:** Agent 2 evaluates their neighbors (Agents 1 and 3). Their favorite available item among neighbors is h_1. Agent 2 points to Agent 1. Since Agent 1 is not in the stack, Agent 1 is pushed. $S = [2, 1]$.
4. **Agent 1 Points:** Agent 1 evaluates their only neighbor (Agent 2). While Agent 1 desires h_3 most globally, h_3 is not held by a neighbor. Their favorite *available* neighbor item is h_2. Agent 1 points to Agent 2.

5. **Leave Phase:** Agent 2 is already in the stack! A trading cycle $C = \{1, 2\}$ is formed. Agents 1 and 2 are popped from the stack, allocated h_2 and h_1 respectively, and **leave** the market.

6. **Share Phase:** The mechanism attempts to share the remaining neighbors of the departed cycle. However, since Agent 1 and Agent 2 both left, Agent 3 is left entirely isolated with no remaining neighbors to share.

7. **Termination:** Following $\mathcal{P}$, Agent 3 is pushed to the empty stack. Having no neighbors, Agent 3 points to themselves, forming a self-cycle. Agent 3 receives h_3 and leaves. The mechanism terminates.

4. The Inefficiency The LS mechanism terminates with the allocation $\mu_{\text{LS}} = (h_2, h_1, h_3)$. However, consider the alternative allocation $\mu_{\text{Global}} = (h_3, h_1, h_2)$, which represents a full cycle where $1 \to 3 \to 2 \to 1$.

- Agent 1 receives h_3 in μ_{Global} vs. h_2 in μ_{LS} (**Strictly better off**).
- Agent 2 receives h_1 in both scenarios (**Equally well off**).
- Agent 3 receives h_2 in μ_{Global} vs. h_3 in μ_{LS} (**Strictly better off**).

Moving from μ_{LS} to μ_{Global} strictly improves the welfare of Agents 1 and 3 without harming Agent 2. Therefore, μ_{LS} is strictly Pareto dominated. Because LS allowed Agents 1 and 2 to form a local cycle and leave *before* the dynamic "Share" phase could connect Agent 1 and Agent 3, the globally optimal cycle was permanently destroyed. This rigorously proves that Global PE cannot be universally guaranteed, necessitating the relaxation to Pairwise Efficiency.

The Cost of Global Pareto Efficiency: The Need for Centralization

It is natural to ask whether a mechanism could be designed to capture the optimal global cycle (e.g., $1 \to 3 \to 2 \to 1$) that the Leave and Share (LS) mechanism missed in the previous example. Achieving this requires abandoning decentralized network rules and introducing a central authority.

As established in Section 8 via the **Mediator-Based Global Cycle Mechanism**, if a central dictator is permitted to view the entire market, ignore network edges, and enforce simultaneous multi-way swaps across distant nodes, Global Pareto Efficiency can be strictly achieved. The mediator would simply observe the global preferences, prevent Agents 1 and 2 from executing their local trade prematurely, and force the 3-way cycle.

However, as dictated by our Impossibility Result (Theorem 7), achieving this Global PE comes at a severe theoretical cost:

- **Violation of Locality:** The trade requires Agents 1 and 3 to exchange endowments despite having no social or network connection, defeating the premise of a socially constrained market.
- **Violation of Strategy-Proofness (SP):** If agents anticipate a global mediator step, they are heavily incentivized to misreport their preferences early on. An agent might strategically claim a highly desired house to use as a "strategic endowment" in the global cycle, destroying the dominant strategy of truth-telling.

Ultimately, the LS mechanism's failure to achieve Global PE is not a flaw in the algorithm, but rather the mathematical price paid to maintain a decentralized, Strategy-Proof, and edge-respecting protocol. By settling for dynamic Pairwise Efficiency, LS strikes the optimal balance between incentive compatibility and network constraints.

The Algorithmic Catch-22: Why Neighbor Manipulation Cannot Force Global PE

Building upon the structural limits of the Leave and Share (LS) mechanism, we must address a subtle behavioral relaxation: What if agents are strictly required to report their *true* preferences, but are allowed to perfectly coordinate, persuade, or "manipulate" their neighbors' pointing actions to stall or orchestrate a trade? Could this type of neighbor manipulation achieve Global Pareto Efficiency?

The answer remains **no**. The failure of LS to achieve Global Pareto Efficiency is immune to neighbor manipulation because the limitation is hard-coded into the graph-theoretic execution rules of the algorithm, not the behavioral choices of the agents.

Consider again the optimal global cycle $C_{\text{Global}} = \{1, 3, 2\}$ on the $1-2-3$ tree network. For this cycle to execute, the mechanism must recognize a closed loop of valid pointers: $1 \to 3 \to 2 \to 1$.

Even if the agents perfectly manipulate each other to cooperate and stall for this exact cycle, they cannot bypass the algorithm's execution sequence, which creates an irreconcilable "Catch-22":

1. **The Edge Constraint:** Agent 1 desires h_3 and must point to Agent 3. However, the LS algorithm strictly prohibits Agent 1 from pointing to Agent 3 unless the edge $(1, 3)$ formally exists in the current active graph.

2. **The Trigger Constraint:** The algorithm only generates the edge $(1, 3)$ during the "Share" phase. This phase is strictly triggered *if and only if* their mutual neighbor, Agent 2, forms a cycle and permanently **leaves** the market.

3. **The Participant Constraint:** For Agent 2 to be part of the multi-way cycle $\{1, 3, 2\}$ and receive their globally optimal house (h_1), Agent 2 must be actively present in the market at the exact moment the cycle forms.

The Algorithmic Paradox: To include Agent 2 in the multi-way trade, Agent 2 must remain active in the market. But to connect Agent 1 and Agent 3 so the multi-way trade can physically form, Agent 2 must permanently leave the market.

Because the algorithm's "Share" phase is strictly *post-mortem* (occurring only after an intermediary departs), an agent can never serve as both a structural bridge to create a new edge and an active participant in the trade that utilizes that new edge. Therefore, even with perfect coordination and neighbor manipulation under truthful reporting, the LS algorithm physically prohibits the execution of the globally optimal cycle.

The Impossibility of Global Pareto Efficiency

While the dynamic "Share" phase resolves specific local bottlenecks, we must acknowledge a fundamental limitation in decentralized diffusion mechanisms.

Theorem 7 (Impossibility of Global PE). *For any tree-structured social network (TSSN) containing three or more agents, no decentralized matching mechanism can simultaneously satisfy Global Pareto Efficiency (PE), Strategy-Proofness (SP), and Individual Rationality (IR) for all preference profiles.*

Proved this theorem earlier.

Because agents only reveal their preferences and neighbor sets sequentially, a globally Pareto Optimal sequence of trades cannot be universally guaranteed without either risking strategic misreporting (violating SP) or forcing an agent into a strictly worse allocation to facilitate a broader cycle (violating IR). Since Global PE is mathematically unachievable under strict mechanism design constraints across all general networks, we must relax our universal optimality metric to *Weak Pairwise Efficiency.*

Achieving Weak Pairwise Efficiency

Because an allocation is a static outcome independent of the mechanism used to produce it, we first define this efficiency metric relative to a specific, given network graph.

Definition 13 (Weak Pairwise Efficiency). Given a network graph $G = (N, E)$, an allocation π is *weakly pairwise efficient* with respect to the edge set E if there does not exist any pair of connected agents $\{i, j\} \in E$ such that $\pi_j \succ_i \pi_i$ and $\pi_i \succ_j \pi_j$.

When evaluating a dynamic protocol such as the Leave and Share mechanism, we assess its final allocation against the *realized* network topology. We state that the mechanism guarantees Weak Pairwise Efficiency by evaluating it against E_{realized}, defined as the union of all edges that existed at any point during the mechanism's execution.

Theorem 8. *A dynamically updated decentralized trading mechanism utilizing the Leave and Share (LS) principle guarantees Weak Pairwise Efficiency.*

Proof. Suppose, for the sake of contradiction, that the LS mechanism terminates and yields an allocation π that is not Weakly Pairwise Efficient. This implies there exists a blocking pair of agents, i and j, who were connected by an edge in E_{realized} at some point during the execution, and who strictly prefer each other's final allocated items: $\pi_j \succ_i \pi_i$ and $\pi_i \succ_j \pi_j$.

Because all agents eventually form a cycle and leave the market, every agent is assigned their final item in a specific round. Without loss of generality, assume agent i forms a cycle and leaves the market in round R_i, and agent j leaves in round R_j, such that $R_i \leq R_j$.

Consider the state of the market in round R_i:

1. Because $R_i \leq R_j$, agent j is still active in the market (or leaving in the exact same cycle).
2. Furthermore, the item agent j will eventually receive, π_j, must also still be available in the market. (If π_j belonged to an agent who left before R_i, agent j could not possibly receive it in round R_j).
3. Because i and j are connected, agent i has a valid path to point to the owner of π_j.

Under the rules of the LS mechanism, agent i points to the agent holding their *most preferred available item* within their network reach. In round R_i, agent i chose to point toward the owner of π_i.

Since π_j was fully available to agent i, but agent i actively chose π_i instead, it must mathematically hold that $\pi_i \succeq_i \pi_j$.

This directly contradicts our initial assumption that agent i strictly prefers agent j's item ($\pi_j \succ_i \pi_i$). Because a mutual block requires strict preference from both parties, the blocking pair cannot exist. Thus, the LS mechanism strictly satisfies Weak Pairwise Efficiency. $\square$

9.5 Demonstrating Theorem 8: Weak Pairwise Efficiency in a 4-Agent Network

To further illustrate the robust guarantees of Theorem 8, we trace the Leave and Share (LS) mechanism on a 4-agent network. This example demonstrates how the dynamic "Share" phase prevents Weak Pairwise Efficiency violations by unlocking complex, multi-agent trading cycles that static mechanisms would block.

1. The Network Structure and Priority Ordering Consider a star graph with 4 agents, where Agent 2 acts as the central hub (the root).

- **Root:** Agent 2 (Distance 0)
- **Leaf Nodes:** Agents 1, 3, and 4 (Distance 1)
- **Initial Edges:** $(1, 2), (2, 3), (2, 4)$

Agents 1, 3, and 4 are completely isolated from each other initially. Following the shortest-path logic of LS, we establish the priority ordering as: $\mathcal{P} = (2, 1, 3, 4)$.

2. Market Setup We define a preference profile where the central Agent 2 is uncooperative, but the outer agents have preferences that form a perfect 3-way cycle.

- **Endowments:** $\mu_0 = (h_1, h_2, h_3, h_4)$
- **True Preferences:**

$$\begin{aligned}
\succ_1 &: h_3 \succ_1 h_1 \succ_1 h_2 \succ_1 h_4 \\
\succ_2 &: h_2 \succ_2 h_1 \succ_2 h_3 \succ_2 h_4 \\
\succ_3 &: h_4 \succ_3 h_3 \succ_3 h_1 \succ_3 h_2 \\
\succ_4 &: h_1 \succ_4 h_4 \succ_4 h_2 \succ_4 h_3
\end{aligned}$$

3. Formal Execution under LS The mechanism processes trades using the stack S based on $\mathcal{P}$:

1. **Initialize:** Stack S is empty.
2. **Push Root:** Agent 2 is pushed into the stack. $S = [2]$.
3. **Round 1 (Leave):** Agent 2 evaluates their neighbors $\{1, 3, 4\}$. However, Agent 2's absolute favorite item is their own endowment, h_2. Agent 2 points to themselves, forming a self-cycle $C = \{2\}$. Agent 2 receives h_2 and **leaves** the market.
4. **Round 1 (Share):** The departed Agent 2 had three active neighbors: Agents 1, 3, and 4. According to the LS rules, all remaining neighbors of the departed agent become fully connected. **Three new edges are dynamically created:** $(1, 3), (3, 4)$, and $(1, 4)$.
5. **Round 2:** The stack is empty. The next agent in $\mathcal{P}$ is Agent 1. Push 1. $S = [1]$.
6. **3-Way Cycle Formation:**
   - Agent 1's favorite available item is h_3. Because Agent 3 is now a direct neighbor, Agent 1 points to Agent 3. Push 3. $S = [1, 3]$.
   - Agent 3's favorite available item is h_4. Because Agent 4 is now a direct neighbor, Agent 3 points to Agent 4. Push 4. $S = [1, 3, 4]$.
   - Agent 4's favorite available item is h_1. Because Agent 1 is now a direct neighbor, Agent 4 points to Agent 1.
7. **Final Execution:** A trading cycle $C = \{1, 3, 4\}$ is formed. Agent 1 receives h_3, Agent 3 receives h_4, and Agent 4 receives h_1. All agents leave, and the mechanism terminates.

4. Verification of Weak Pairwise Efficiency The LS mechanism terminates with the allocation $\mu_{\text{LS}} = (h_3, h_2, h_4, h_1)$. To verify Weak Pairwise Efficiency (Theorem 8), we examine the realized edge set E_{realized}, which now includes the initial star edges plus the dynamically created complete triangle: $\{(1, 2), (2, 3), (2, 4), (1, 3), (3, 4), (1, 4)\}$.

We check for any pair of connected agents who mutually prefer each other's final items:

- **Agent 1** received h_3 (their #1 top choice). They will never agree to a swap.
- **Agent 2** received h_2 (their #1 top choice). They will never agree to a swap.
- **Agent 3** received h_4 (their #1 top choice). They will never agree to a swap.
- **Agent 4** received h_1 (their #1 top choice). They will never agree to a swap.

Because the dynamic "Share" phase successfully recognized that the bottleneck (Agent 2) was leaving, it instantly connected the outer network. This allowed Agents 1, 3, and 4 to execute a 3-way global trade that perfectly optimized their welfare. Consequently, zero mutually beneficial pairwise swaps exist in the final graph, flawlessly maintaining the Weak Pairwise Efficiency guarantee outlined in Theorem 8.

Comparison with SWN and YRMH-IGYT

It is important to contextualize this Weak Pairwise Efficiency guarantee against static network mechanisms like Swap With Neighbors (SWN) and You Request My House - I Get Your Turn (YRMH-IGYT).

Standard SWN and static network-restricted YRMH-IGYT *do* satisfy Weak Pairwise Efficiency, but strictly relative to their *initial* static edges. However, because they cannot bypass intermediaries, they dramatically fail to achieve **Weak Pairwise Efficiency** when evaluated against dynamically expanded network topologies.

Consider again our bottleneck line graph: $1-2-3$. Under SWN, Agent 1 and Agent 3 mutually prefer each other's items ($h_3 \succ_1 h_1$ and $h_1 \succ_3 h_3$). They form a textbook blocking pair in a global sense. However, because they do not share an initial edge, SWN does not recognize them as a valid pairwise block. When Agent 2 refuses to act as an intermediary, SWN terminates, leaving a globally inefficient allocation.

The brilliance of the "Leave and Share" (LS) algorithm is that it dynamically expands the definition of who a "neighbor" is. By connecting Agent 1 and Agent 3 after Agent 2 leaves, LS expands the edge set E_{realized}. By guaranteeing Weak Pairwise Efficiency over this much larger, dynamically realized graph, LS eliminates hidden blocking pairs that SWN and static YRMH-IGYT leave trapped behind uncooperative intermediaries.

10 Conclusion

This project establishes that while standard mechanisms satisfy strong incentive and stability properties under tree-structured social networks (TSSNs), they inherently fail to guarantee global efficiency. The final conclusions of our theoretical and computational analysis center on the fundamental trade-offs between incentives, stability, and efficiency in network-constrained matching markets:

- **Failure of Global Efficiency in Standard Mechanisms:** While the YRMH-IGYT mechanism satisfies desirable properties like strategy-proofness and strict core stability for neighbors (SC4N), it fundamentally fails to guarantee global pair efficiency and Pareto efficiency. Simulations demonstrate that this inefficiency scales with market size, making Pareto-inefficient allocations almost certain in large networks.
- **The Impossibility Result:** We mathematically prove that it is impossible for any mechanism on a TSSN to simultaneously satisfy strategy-proofness, individual rationality, SC4N, and pair efficiency.

- **The Cost of Pareto Efficiency:** To achieve true Pareto efficiency, strategy-proofness and network constraints must be relaxed. For example, a mediator-based global cycle mechanism can achieve Pareto efficiency, but it does so by allowing trades across distant agents, thus sacrificing both network locality and strategy-proofness.
- **Limitations of Static Local Protocols:** The intuitive Swap with Neighbors (SWN) mechanism guarantees local stability and individual rationality but is highly vulnerable to strategic manipulation. Furthermore, SWN fails global efficiency because uncooperative intermediary agents act as bottlenecks, trapping the network in suboptimal allocations.
- **Success of Dynamic Network Updating:** To overcome the limitations of static networks and intermediary bottlenecks, the dynamic "Leave and Share" (LS) mechanism is highly effective. By removing agents once they trade and dynamically connecting their remaining neighbors, LS successfully restores strategy-proofness and strictly guarantees **Weak Pairwise Efficiency**.
- **The Ultimate Trade-off:** The central takeaway of this project is that in decentralized, network-constrained matching environments, achieving global efficiency fundamentally requires relaxing either incentive constraints (strategy-proofness), the locality of the network, or local stability. Understanding, managing, and balancing these trade-offs remains the primary challenge in modern matching theory.

11 Future Research Directions

This project establishes that static mechanisms, such as YRMH-IGYT and SWN, satisfy strong incentive and stability properties under tree-structured social networks but fundamentally fail to guarantee global efficiency due to intermediary bottlenecks. However, our analysis of the Leave and Share (LS) mechanism demonstrates that dynamic network updating offers a promising pathway to bypass these limitations and restore **Weak Pairwise Efficiency**.

Several important directions arise from this work:

1. Characterization of Efficient Mechanisms on TSSNs

A fundamental open question is to characterize the entire class of mechanisms that achieve Pareto efficiency under network constraints. In particular, it would be valuable to identify the minimal conditions under which efficiency can be restored without fully sacrificing strategy-proofness.

2. Trade-off Between Locality and Efficiency

Our impossibility result shows that strictly local, static mechanisms cannot guarantee Pareto efficiency. A deeper theoretical analysis of the trade-off between locality, stability, and efficiency could lead to sharp characterization theorems.

3. Generalizing Dynamic Network Topologies

The success of the Leave and Share (LS) mechanism highlights the power of dynamic graph updating. Future work should formalize the "Share" principle across different network topologies. Investigating alternative rules for how edges are inherited or created when agents leave the market could yield mechanisms with even stronger global efficiency guarantees.

4. Strategic Complexity in Dynamic Mechanisms

While LS maintains Strategy-Proofness in our evaluated scenarios, dynamic networks introduce new strategic dimensions. If agents know the network will update in subsequent rounds, they might attempt complex temporal manipulations (e.g., waiting to trade until a better neighbor is shared). A rigorous analysis of dynamic incentive compatibility in sequential matching markets is a crucial next step.

5. Minimal Relaxation Mechanisms

The sibling swap mechanism improves efficiency only partially within static constraints. Designing hybrid mechanisms that combine minimal local relaxations (like sibling swaps) with limited dynamic updating (like LS) while preserving partial stability is an interesting direction.

6. Randomized Mechanisms

Randomized mechanisms may provide improved efficiency in expectation. Extending random priority or random cycle-based mechanisms to both static and dynamic network settings could yield better welfare guarantees.

7. Large-Market Asymptotics

Empirical results suggest that the probability of inefficiency approaches one in static networks as the number of agents grows. A rigorous asymptotic analysis of

$$\Pr(\text{Pareto inefficiency})$$

under both static and dynamic mechanisms would provide deeper insights into large-market behavior and the scaling limits of local trading protocols.

8. Beyond Tree Structures

Extending the analysis of both static mechanisms (YRMH-IGYT) and dynamic mechanisms (LS) to general networks, including directed acyclic graphs and arbitrary social networks, may reveal stronger impossibility results and richer structural phenomena.

9. Preference Restrictions

While single-peaked preferences introduce structure, our results indicate that they do not eliminate inefficiency under network constraints. Identifying specific preference domains that naturally mitigate intermediary bottlenecks remains an open problem.

Overall, the results highlight that achieving efficiency in network-constrained environments requires relaxing either incentive constraints, static network locality, or absolute stability. Understanding these trade-offs—and leveraging dynamic topology updates to navigate them—remains a central frontier in matching theory.

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